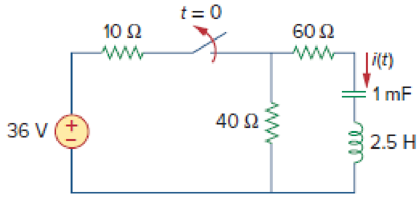
Problem

Find *i*(*t*) for *t* > 0 in the circuit of Fig. 16.43.

Figure 16.43:



Step-by-step solution

Step 1 of 5

Refer to Figure 16.43 in the textbook.

The switch is initially closed for $t<0$. So, the circuit remains in the steady state for $t<0$ and the inductor and the capacitor are replaced by the short and open circuit respectively.

Redraw the circuit for $t=0^{-}$ as shown in Figure 1.

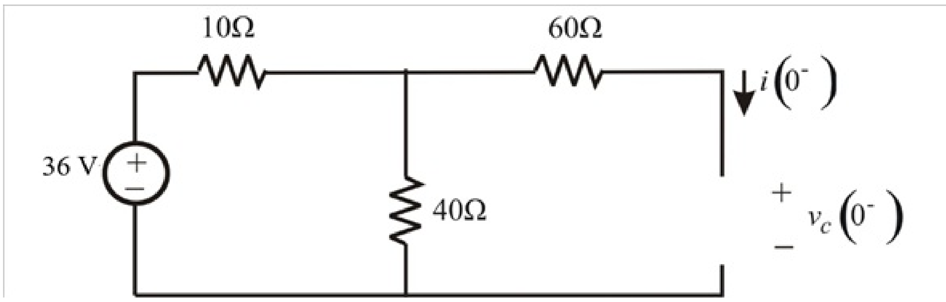


Figure 1

Step 2 of 5

Calculate the initial voltage across the capacitor using voltage division method.

$$\begin{aligned} v_c\left(0^{-}\right) &= 36\left(\frac{40}{40+10}\right) \\ &= 36\left(\frac{4}{5}\right) \\ &= 28.8 \text{ V} \end{aligned}$$

Since the circuit is open, the initial inductor current is $i\left(0^{-}\right)=0 \text{ A}$.

In steady state, $i_L\left(0^{-}\right)=i_L\left(0^{+}\right)=i_L(0)$ and $v_c\left(0^{-}\right)=v_c\left(0^{+}\right)=v_c(0)$.

So,

$$i(0)=0 \text{ A}$$
$$v_c(0)= 16 \text{ V}$$

Step 3 of 5

Redraw the circuit for $t>0$ in s-domain as shown in Figure 2.

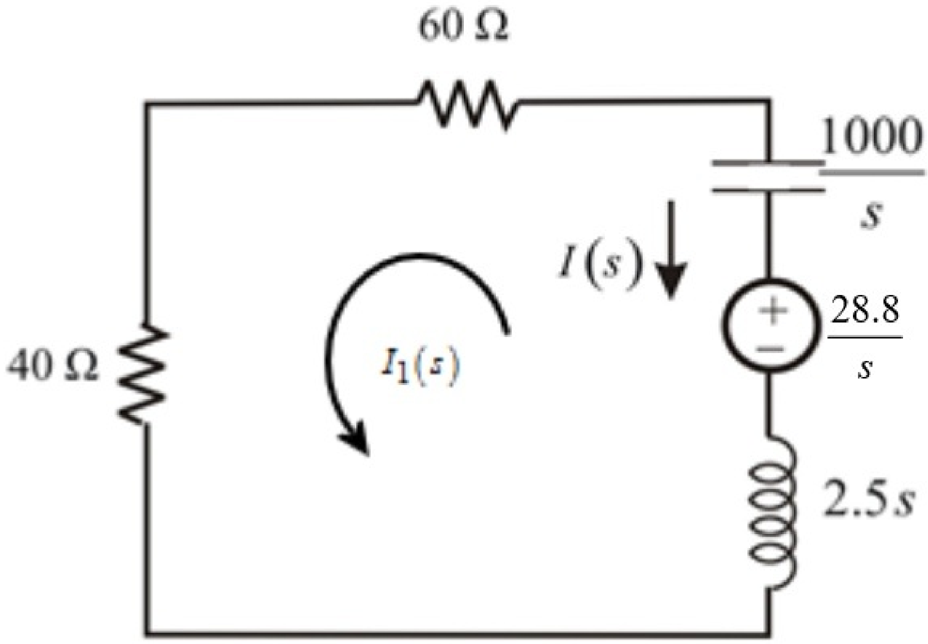


Figure 2

Step 4 of 5

Apply Kirchhoff's voltage law for the loop $I_1(s)$.

$$-\frac{28.8}{s}+2.5sI_1(s)+\frac{1000}{s}I_1(s)+40I_1(s)+60I_1(s)=0$$

$$I_1(s)\left[\frac{2.5s^2+1000+100s}{s}\right]=\frac{28.8}{s}$$

$$I_1(s)\left[s^2+40s+400\right]=11.52$$

$$I_1(s)=\frac{11.52}{s^2+40s+400}$$

$$=\frac{11.52}{(s+20)^2}$$

Step 5 of 5

Apply inverse Laplace transform on both sides of equation $I_1(s)$.

$$i_1(t)=11.52te^{-20t}u(t)\text{ A}$$

Calculate $i(t)$.

$$i(t)=-i_1(t)$$
$$=-11.52te^{-20t}u(t)\text{ A}$$

Therefore, the current flowing through the inductor $i(t)$ is $\boxed{-11.52te^{-20t}u(t)\text{ A}}$.